

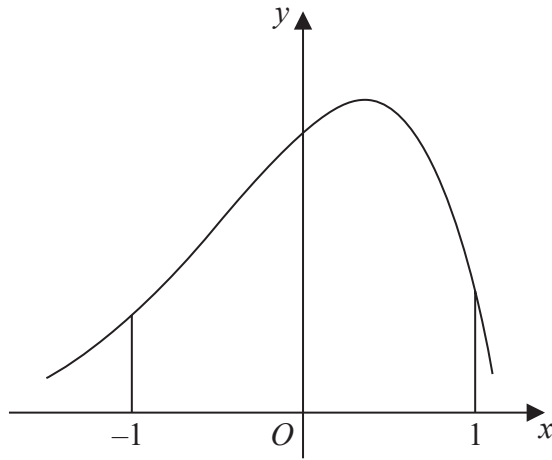


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$
 The table below shows some corresponding values of x and y for this curve.
 The values of y are given to 3 decimal places.

x	-1	-0.5	0	0.5	1
y	2.287	4.470	6.719	7.291	2.834

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for

$$\int_{-1}^1 f(x) \, dx$$

giving your answer to 2 decimal places.

(3)

(b) Use your answer to part (a) to estimate

(i) $\int_{-1}^1 (f(x) - 2) \, dx$

(ii) $\int_1^3 f(x - 2) \, dx$

(3)